

ASWIKA NADHEESWARAN

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A tenacious ECE student with a strong interest in C programming and embedded software development. With a foundation in C, programming fundamentals, and problem-solving, along with hands-on experience through technical projects, I am seeking an entry-level opportunity to apply my technical knowledge, develop my software and debugging skills, and contribute to building reliable and efficient software solutions.

PROFESSIONAL EXPERIENCE

Systems Integration Intern | Yudu Robotics June 2026 – Present

- Contributing to the product development cycle by designing and integrating LEGO-based robotic systems.
- Designing comprehensive educational curricula to pair with the robotics hardware for interactive learning.

Founding Engineer | eNowatiX Technologies (June 2024 - March 2026)

- Engineered and prototyped microcontroller-based educational hardware, managing the end-to-end embedded development cycle from component selection to final system deployment.
- Guided 100+ students in building functional hardware prototypes, teaching core concepts in sensor integration, basic firmware, and embedded electronics.
- Secured ₹17,000 in funding by successfully pitching scalable hardware designs and embedded solutions at technical competitions across Kerala, Trichy, Salem, and Coimbatore.

PATENTS & INTELLECTUAL PROPERTY

Modular Microcontroller with Removable Wireless Communication Module
Status: Published on 27/03/2026
Application No: 202641031100
Impact: Designed a plug-and-play hardware architecture tailored for STEM education, allowing for easily swappable communication protocols (Wi-Fi, Bluetooth) to reduce prototyping costs and maximize learning modularity.

Gas Leakage and Adulteration Detection Device and a Method
Status: Filed
Application No: 202641094563
Impact: Designed an IoT-integrated dual-sensor (MQ-4/MQ-6) framework for real-time LPG leakage and fuel adulteration monitoring. The automated system instantly detects unauthorized hydrocarbons like methane and CNG, triggering multi-channel alerts to enhance household safety and promote supplier accountability with minimal human intervention

TECHNICAL PROFILE & CORE COMPETENCIES

Programming Languages: C
Operating Systems: Ubuntu Linux (Basic)
Embedded Development: Arduino IDE, MATLAB & Simulink, Wokwi, Tinkercad
Microcontrollers & IoT: ESP32, ESP8266, Arduino (AVR)
Hardware Design & Simulation: OrCAD (PCB Design), Vivado, Quartus Prime, LTspice
Areas of Interest: Embedded Systems, Embedded Linux, Firmware Development, IoT, Real-Time Systems, PCB Design, Electronic Product Development, VLSI

INTERNSHIP AND TRAINING

Wizaard System: 7-day hands-on PCB Design training (Dec 2024 – Jan 2025).
BSNL Salem: 5-day training on Fundamentals of Telecommunication (July 2024).
Learn Flu: 2-month VLSI specialization course (May – July 2024).
SLM InfoTech: 7-day hands-on internship in Basic Electronics (Feb 2024).

HACKATHON EXPERIENCE

TN-IMPACT 2026: Secured a specialized internship for a real-time firmware implementation of an Air Pollution Monitoring & Filtering System, focusing on hardware-software synchronization.
Cognizant Technoverse Hackathon 2026: Shortlisted for the Agent Builder Challenge (Final Round), demonstrating expertise in complex logic development and system architecture.
Forest Department Hackathon (National Level): Selected among the Top 12 teams nationally and awarded ₹10,000 for engineering embedded firmware solutions for human-wildlife coexistence at Dehradun.
SIH Hackathon 2025: Contributed to the Smart Electric Fence project with a primary focus on iterative design and system optimization for real-time safety protocols.
StartupTN Hackathons (Olimayam 2.0 & Women Hackathon): Won 2 Smart Cards for technical innovation in STEM education kits, highlighting skills in rapid hardware prototyping and teamwork.
SIH Hackathon 2024: Implemented multi-sensor integration and high-speed firmware prototyping for an automated Smart Dustbin project.

TECHNICAL PROJECTS

- **Smart Electric Fence:** Built an Arduino power monitor (ACS712, LM393) to track I/O metrics and detect power leakage.
- **Air Pollution System:** Developed an IoT emission monitor using C logic and MQ-2 sensors, achieving a <2s response time.
- **Smart Dustbin:** Engineered a waste segregation prototype using sensor fusion (IR, moisture) and servos for 95% sorting accuracy.
- **Fire Fighting Robot:** Created a Bluetooth-controlled Arduino robot with IR flame detection, obstacle avoidance, and 85% pump accuracy.
- **Bluetooth-Controlled Robot:** Optimized an Arduino robot featuring HC-05 smartphone control (30m range) and reliable ultrasonic collision avoidance.
- **Multi-Sensor Vitals Identifier:** Prototyped a real-time healthcare device integrating blood group, SpO2, heart rate, and fingerprint sensors.
- **Human & Object Detection:** Designed a spatial awareness system using PIR and ultrasonic sensors for precise presence and proximity tracking.
- **Digital Hardware Calculator:** Programmed an embedded Arduino calculator (LCD, matrix keypad) using optimized C logic for instant arithmetic.

KEY ACHIEVEMENTS

Innovation & Pitching Success: National Finalist in MSME 4.0 & 5.0 and TANSEED 7.0 (Round 2). Consistently recognized for Idea Pitching, securing cash awards at the National Forest Department Hackathon (₹10,000, Top 12), Dakshaa 2025 (₹10,000, 1st Place), KIOT (₹5,000), Karpagam College (₹1,000), Sona College (₹600), and Muthayammal College (₹300). 1st Prize at Lead College Kerala.
Leadership & Volunteering: MoE Innovation Ambassador (Organized 30 iStart events at KIOT); NSS Member (Organized 5 community events); Active IEEE Member.
Awards: Achievers Award at KIOT Achievers Day (2026 & 2025); Best Student in Extra-Curricular Activities at KIOT Annual Day (2025).

EDUCATION

Knowledge Institute of Technology, Salem — CGPA (5th Semester): 8.38
B.E. Electronics and Communication Engineering — **2023 - 2027**

Salem Sri Sowdeswari Matric and Hr Sec School
Class 12th — **2021 - 2023**
Class 10th — Percentage: 85.17%

ADDITIONAL INFORMATION

Languages: English, Tamil, Kannada.